

tf.compat.v1.math.reduce_variance Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/math/reduce_variance

Computes the variance of elements across dimensions of a tensor.

Function Signatures

```
tf.compat.v1.math.reduce_variance( input_tensor, axis=None,
keepdims=False, name=None )
```

Code Examples

```
tf.compat.v1.math.reduce_variance(  
input_tensor, axis=None, keepdims=False, name=None  
)
```

```
tf.compat.v1.math.reduce_variance(  
input_tensor, axis=None, keepdims=False, name=None  
)
```

```
x = tf.constant([[1., 2.], [3., 4.]])  
tf.math.reduce_variance(x)  
  
tf.math.reduce_variance(x, 0)  
  
tf.math.reduce_variance(x, 1)
```

```
x = tf.constant([[1., 2.], [3., 4.]])
```

```
tf.math.reduce_variance(x)
```

```
tf.math.reduce_variance(x, 0)
```

Documentation

Computes the variance of elements across dimensions of a tensor.

Reduces `input_tensor` along the dimensions given in `axis`.

Unless `keepdims` is true, the rank of the tensor is reduced by 1 for each of the entries in `axis`, which must be unique. If `keepdims` is true, the reduced dimensions are retained with length 1.

If `axis` is `None`, all dimensions are reduced, and a tensor with a single element is returned.

For example:

`## Returns`

numpy compatibility

Equivalent to `np.var`

Please note `np.var` has a `dtype` parameter that could be used to specify the output type. By default this is `dtype=float64`. On the other hand, `tf.math.reduce_variance` has aggressive type inference from `input_tensor`.